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New Approximation Bounds for Small-Set Vertex Expansion

Suprovat Ghoshal[*] Anand Louis[†]

Abstract

The vertex expansion of graph is a fundamental graph parameter. Given a graph $G = (V, E)$ and a parameter $\delta \in (0, 1/2]$, its δ-SSVE is defined as

$$\min_{S:|S|=\delta|V|} \frac{\left|\partial^V(S)\right|}{\min\{|S|, |S^c|\}}$$

where $\partial^V(S)$ is the vertex boundary of a set S. The SSVE problem, in addition to being of independent interest as a natural graph partitioning problem, is also of interest due to its connections to the StrongUniqueGames problem [16]. We give a randomized algorithm running in time $n^{\mathsf{poly}(1/\delta)}$, which outputs a set S of size $\Theta(\delta n)$, having vertex expansion at most

$$\max\left(O(\sqrt{\phi^* \log d \log(1/\delta)}), \tilde{O}(d \log(1/\delta)) \cdot \phi^*\right),$$

where d is the largest vertex degree of the graph, and ϕ^* is the optimal δ-SSVE. The previous best known guarantees for this were the bi-criteria bounds of $\tilde{O}(1/\delta)\sqrt{\phi^* \log d}$ and $\tilde{O}(1/\delta)\phi^*\sqrt{\log n}$ due to Louis-Makarychev [TOC'16].

Our algorithm uses the basic SDP relaxation of the problem augmented with poly$(1/\delta)$ rounds of the Lasserre/SoS hierarchy. Our rounding algorithm is a combination of rounding algorithms of [37, 7]. A key component of our analysis is novel Gaussian rounding lemma for hyperedges which might be of independent interest.

1 Introduction

Graph partitioning problems are a fundamental class of problems that are studied extensively in theory and practice. In theory, they have many connections to metric embeddings [23, 5], Markov chains [31, 3], etc. in addition to connections to fundamental open questions such as the *Unique Games Conjecture* [21] and *Small-set Expansion Hypothesis* [34]. In practice, they are extensively used as inexpensive pre-processing steps for simplifying an optimization problem into isolated sub-problems of smaller size e.g., dynamic algorithms [38], clustering algorithms [20], ranking [14], etc. A commonly studied problem in this context is to design algorithms where the goal is to output a partition which minimizes the *edge expansion* of the set i.e,. given a d-regular graph $G = (V, E)$ on n-vertices, the goal is to minimize

$$\phi^{\mathsf{E}}_G(S) := \frac{|E(S, S^c)|}{d|S|}$$

over all sets of size at most $n/2$, where $\phi^{\mathsf{E}}_G(S)$ is referred to as the edge expansion or "conductance" of the set S. This, and its several variants, have been studied extensively in the literature, on account of being natural optimization problems with deep connections to several areas of mathematics such as *Isoperimetric Inequalities* [2] and *Metric Embedding* [23]. Of particular interest in this setting is the problem of minimizing the δ-small set edge expansion of a graph – denoted by ϕ^{E}_δ – where given a parameter $\delta \in (0, 1/2]$, the goal is to compute a set with the minimum edge expansion over all sets of size δn. Introduced by Raghavendra and Steurer [34], the SmallSetEdgeExpansion problem (SSE in short) is central to the *Small-set Expansion Hypothesis* (SSEH) and is closely related to Khot's Unique Games Conjecture [21].

[*]Northwestern University & TTIC
[†]Indian Institute of Science, Bangalore

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The focus of this work is the related problem of minimizing *vertex expansion* ϕ^{V} in graphs. Formally, given a graph $G = (V, E)$, the vertex expansion of a set $S \subseteq V$ is defined as

$$\phi^{\mathsf{V}}_G(S) \stackrel{\text{def}}{=} \frac{\left|\partial^V_G(S)\right|}{|S|}$$

where $\partial^V_G(S)$ is the vertex boundary[1] of the set S. The question of finding sets with the minimum vertex expansion has been studied by several works in the past – the best known upper bounds are by [13] who gave an $O(\sqrt{\log n})$-approximation algorithm, and by [30], who gave a $O(\sqrt{\phi^{\mathsf{V}} \log d})$ bound for approximating vertex expansion on graphs of maximum degree d; they also gave a matching (up to constant factors) lower bound based on SSEH.

Here we study the "small-set" variant of the above, namely the SmallSetVertexExpansion (SSVE) problem. Formally, given a parameter $\delta \in (0, 1/2]$, the δ-SSVE of G is defined as

$$\phi^{\mathsf{V}}_\delta \stackrel{\text{def}}{=} \min_{S:|S|=\delta|V|} \phi^{\mathsf{V}}(S).$$

The δ-small set vertex expansion of a graph, denoted by ϕ^V_δ henceforth, while being an independently interesting quantity on its own as a generalization of edge expansion, is also useful for studying hypergraph analogues of graph partitioning primitives such as Sparsest Cut and Cheeger's Inequality [10, 28]. Furthermore, minimizing the (small set) vertex expansion appears to be a key combinatorial bottleneck towards approximating CSPs with hard constraints such as the StrongUniqueGames problem[22, 16] – similar to the well established connection between UniqueGames and SSE [33, 36].

However, while ϕ^{V}_δ is a natural variant of edge expansion, as a measure of connectivity, its behavior can be quite different from the edge expansion counterpart ϕ^{E}_δ. In particular, unlike edge expansion, the vertex expansion of a set does not immediately admit a analogous re-interpretation in terms of random walks. In fact, note that it is entirely possible for a graph to have ϕ^{V}_δ much larger than 1 – for e.g., it is easy to show that for all constant δ, d-regular random graphs can have ϕ^V_δ as large as $\Omega(d)$. As a result, the techniques and results for approximating ϕ^{E}_δ are unlikely to apply as is to the setting of ϕ^{V}_δ. Furthermore, it is expected that the approximation curve[2] of ϕ^{V}_δ also depends on the maximum degree d [30] in addition to parameters δ and ϕ^{V}_δ. To that end, correctly characterizing the three-way trade-off between the parameters $\phi^{\mathsf{V}}_\delta, \delta$ and d in the approximation curve has been challenging, leading us to question:

(1.1) $\quad\triangleright$ *What is the right form of the approximation curve of* SSVE *as a function of* $\phi^{\mathsf{V}}_\delta, \delta$ *and* d*?*

Towards this, a couple of previous works combined with some additional observations paint a partial yet somewhat helpful picture. Firstly, there have been few works which make progress in terms of upper bounds: [28] gives an algorithm which outputs $O(\delta n)$-sized sets with vertex expansion at most $\tilde{O}((1/\delta)) \cdot \sqrt{\phi^{\mathsf{V}}_\delta \log d}$. Moreover, by combining a straightforward reduction from edge expansion to vertex expansion with the bounds from [35], one can derive an algorithm which outputs a $O(\delta n)$-sized set with vertex expansion at most $O(d) \cdot \sqrt{\phi^V_\delta \log(1/\delta)}$. On the other hand, one can guess that for certain ranges of parameters $\phi^{\mathsf{V}}_\delta, d$ and δ, the right approximation might be the (analytic) vertex expansion of $(1-\epsilon)$-noisy Gaussian graphs[3] – this is known (for e.g., [16]) to be $\alpha_{\text{SSE}} := \sqrt{\epsilon \log(d) \log(1/\delta)}$. Finally, [28] also showed that the basic SDP relaxation admits an integrality gap of $\alpha_{\text{Int}} := \Omega(\min\{d, 1/\delta\})$.

1.1 Our Results

This paper makes progress towards giving a more unified answer to (1.1) by giving new approximation bounds, as well as strengthening the existing unconditional hardness, for the δ-SSVE problem. Our first main result is the following new approximation bound for the SSVE problem:

[1] Formally, the vertex boundary of a set S is defined as $\partial^V_G(S) := \{u \in S^c | \exists v \in S, (u, v) \in E\}$.

[2] By approximation curve, we mean the mapping $\epsilon \to s_\delta(\epsilon)$, which maps ϵ to the best possible approximation guarantee one can hope to efficiently achieve over instances with $\phi^{\mathsf{V}}_\delta \leq \epsilon$.

[3] The $(1-\epsilon)$-noisy Gaussian graph is the infinite graph on $\mathbb{R}^d$ whose random walk is governed by the Ornstein-Uhlenbeck operator [32] with correlation parameter $(1-\epsilon)$.

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THEOREM 1.1. *The following holds for any constant $\delta \in (0, 1/2)$. There exists a randomized algorithm which on input a graph $G = (V, E)$ of maximum degree $d \in \mathbb{N}$, outputs a set of size $\Theta(\delta) \cdot n$ having vertex expansion at most*

$$O\left(\sqrt{\phi_\delta^V \log d \log(1/\delta)}\right) + \phi_\delta^V \cdot O(d)(\log(d)\log(1/\delta))^2.$$

in time $n^{\mathrm{poly}(1/\delta)}$.

The approximation bound from the above theorem can be interpreted to be as $\max\{\alpha_{\mathrm{SSE}}, \tilde{O}(\alpha_{\mathrm{Int}} \cdot \phi_\delta^{\mathsf{V}})\}$ where $\alpha_{\mathrm{SSE}} := \sqrt{\epsilon \log(d)\log(1/\delta)}$ is the (conjectured) SSEH based hardness (see Remark 1), and α_{Int} is the integrality gap of Sum-of-Squares (SoS) lifting of the SDP relaxation (see Proposition 4.1 of full version). We point out that there is a subtle difference between the setting of the above theorem – where the non-expanding set is promised to be of relative size δ, and the algorithm also outputs a set of size $\approx \delta$ – and those of previous related works [35, 28] where the promise and the approximation guarantee are for sets of sizes at most $O(\delta n)$. Nevertheless, we still compare the guarantees of Theorem 1.1 with known bounds.

- Louis and Makarychev [28] give algorithms which output sets of size at most $(1 + \eta)\delta$ with vertex expansion at most $O_\eta(1/\delta) \cdot \sqrt{\phi_\delta^{\mathsf{V}} \log d}$ for any η, with the $O_\eta(\cdot)$ hiding poly$(1/\eta)$-terms. Then for $\delta^{0.99} \leq O(1/\sqrt{\phi_\delta^{\mathsf{V}}}d)$, Theorem 1.1 gives improved approximation guarantees.
- Reducing vertex expansion to edge expansion, and plugging in the guarantees of [35] for SSE yields an algorithm which outputs sets of size at most δn with vertex expansion at most $O(d\sqrt{\phi_\delta^{\mathsf{V}} \log(1/\delta)})$. Again, when $\phi_\delta^V \leq 1/\log^4(1/\delta)$, Theorem 1.1 gives improved approximation guarantees in comparison.

Combining the above immediately yields the following corollary.

COROLLARY 1.1. *Given a graph G with maximum degree d such that $\phi_\delta^{\mathsf{V}} \leq 1/(d^2 \log^4(1/\delta))$, there exists a $n^{\mathrm{poly}(1/\delta)}$-time algorithm that returns a set of size $(1 \pm o(1))\delta n$ with vertex expansion at most $O(\sqrt{\phi_\delta^{\mathsf{V}} \log d \log(1/\delta)})$.*

Lower bounds. We also give conditional and unconditional hardness results which hold for more general settings of ϕ_δ^{V}. The first result shows that it is SSEH hard to get a $f(d)$-approximation for SSVE for any increasing nonnegative function f.

THEOREM 1.2. *Assuming the SSEH, the following holds for any computable nonnegative increasing function $f : \mathbb{N} \to \mathbb{R}_+$. There exists an absolute constant $\epsilon_0 \in (0, 1)$ such that for any $\epsilon \in (0, \epsilon_0)$, there exists $\delta = \delta(\epsilon)$ and a degree d such that given a graph $G = (V, E)$ of maximum degree d it is* NP*-Hard to distinguish between*

- **YES**: $\phi_\delta^V(G) \leq \epsilon$
- **NO**: $\phi_\delta^V(G) \geq f(d) \cdot \epsilon$

REMARK 1. *It is widely conjectured that for graphs with degree d, it should be hard to distinguish between whether $\phi_\delta^V \leq \epsilon$ and $\phi_\delta^V \geq \sqrt{\epsilon \log d \log(1/\delta)}$. This is consistent with known results for δ-small set edge expansion [36], for which it is known that it is hard to distinguish between $\phi_\delta^E \leq \epsilon$ and $\phi_\delta^E \geq \sqrt{\epsilon \log(1/\delta)}$. This is also consistent with known hardness for vertex expansion [30] which states that it is hard to distinguish between the cases $\phi^V \leq \epsilon$ and $\phi^V \geq \sqrt{\epsilon \log d}$.*

Additionally, we also strengthen the integrality gap construction of [28] to show a $\Omega(d)$-integrality gap for the $O(n/\log n)$-round SoS lifting of the basic SDP (Proposition 4.1 in full version).

Additional Algorithmic Applications. Our algorithm for Theorem 1.3 is based on solving a SoS lifting of the strengthened SDP relaxation, followed by a conditioning + Gaussian rounding step. We use this framework

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to derive new approximation guarantees for the related problem of Hyper-SSE. Given a hypergraph $H = (V, E, w)$ with hyperedge weights $w : E \to \mathbb{R}_+$, the expansion of a set $S \subset V$ is defined as

$$\phi^{\mathsf{E}}_H(S) \stackrel{\text{def}}{=} \frac{w\left(\partial^E_H(S)\right)}{\min\left\{\mathsf{Vol}(S), \mathsf{Vol}(S^c)\right\}}, \tag{1.2}$$

where for a subset S, $\phi^{\mathsf{E}}_H(S)$ denotes the hyperedge boundary[4] and $\mathsf{Vol}(S)$ denotes its volume, defined as the sum of the degrees of the vertices in S. The following theorem states our guarantees for HYPER-SSE.

THEOREM 1.3. *There is a randomized algorithm which takes as input a r-uniform hypergraph $H = (V, E, w)$ with hyperedge weights $w : E \to \mathbb{R}^+$, max-degree $d_{\max}$, and containing a set of relative weight δ with expansion ϕ^{E}_δ, runs in time $n^{\text{poly}(1/\delta)}$, and outputs a set of relative weight $\delta(1 \pm o_\delta(1))$ having hyperedge expansion at most*

$$O\left(\sqrt{(d_{\max}/r)\phi^{\mathsf{E}}_\delta \log(1/\delta) \log r} + \tilde{O}(r)\phi^*(\log(1/\delta))^2\right)$$

Again, we compare this with the $\tilde{O}(1/\delta)\sqrt{(d_{\max}/r)\phi^{\mathsf{E}}_\delta \log r}$-bound from [28]; Theorem 1.3 again gives better guarantees in the small volume + small expansion setting i.e., $\delta^{0.99} \leq \cdot d/\sqrt{\phi^{\mathsf{E}}_\delta}$.

1.2 Related Works

Graph/Hypergraph Partitioning Problems. There is a long line of works which study graph/hypergraph partitioning while aiming to minimize various notions of expansion. The simplest variant where the objective is to minimize expansion i.e., SPARSEST CUT has been studied extensively. Leighton and Rao [27] first gave a $O(\log n)$-approximation for it. The breakthrough work of Arora, Rao and Vazirani [6] gave an $O(\sqrt{\log n})$-approximation for this problem. Feige, Hajhiyaghayi and Lee [13] study the balanced vertex separator problem for which they gave a $O(\sqrt{\log n})$-approximation. Louis, Raghavendra and Vempala [30] gave a $O(\sqrt{\phi^{\mathsf{V}} \log d})$ bound for minimizing vertex expansion over graphs with max-degree at most d; they also proved a matching (up to constant factors) lower bound based on SSEH.

Small Set Expansion. The SSE problem was introduced by Raghavendra and Steurer [34] in the context of the Small Set Expansion Hypothesis, in an attempt to characterize the combinatorial structure underlying hard instances of CSPs such as UNIQUE GAMES. Raghavendra, Stuerer and Tetali [35] gave a $O(\sqrt{\phi^{\mathsf{E}}_\delta \log(1/\delta)})$ bound for SSE– this was later matched (up to constant factors) by the work of Raghavendra, Steurer and Tulsiani [36]. In terms of multiplicative approximation, Bansal et al. [9] gave a bi-criteria $O(\sqrt{\log n \log(1/\delta)})$ approximation for SSE. There have also been a sequence of works which show that SSE is tractable on graphs having "low threshold rank", for e.g, see [24], [19]. On other other hand, Arora, Barak and Steurer [4] gave a subexponential time algorithm for SSE on general instances. SSE is also studied in the context of higher order variants of Cheeger's inequality [26, 29] which relate the multi-way expansion with the higher eigenvalues of the normalized Laplacian matrix of the graph. Louis and Makarychev [28] gave $\tilde{O}(1/\delta)\sqrt{\log n}$-approximation algorithms for the δ-SSVE and δ-HYPER-SSE problems respectively. Additionally, for graphs of maximum degree d, they also gave a $\tilde{O}(1/\delta)\sqrt{\phi^{\mathsf{V}}_\delta \log d}$-bound for SSVE.

CSPs with Cardinality Constraints. SSE and HYPER-SSE can also be interpreted as Constraint Satisfaction Problems (CSPs) with global cardinality constraints. CSPs with cardinality constraints model a wide array of combinatorial optimization problems such as MAX-BISECTION, MIN-BISECTION, DENSEST-k-SUBGRAPH, MIN-MAX-PARTITION-etc. The global constraint influences the tractability of underlying CSP. For e.g., Austrin and Stankovic [8] showed that adding cardinality constraints to MAX-CUT makes it strictly harder. Therefore, there have been several attempts to study this class of CSPs on its own. Raghavendra and Tan [37] proposed a general framework for approximating CSPs with cardinality constraints, and used it to give a 0.85-approximation for MAX BISECTION, which was later improved to 0.8776 by Austrin, Bennabas and Georgiou [7]. There have been several works [19, 18, 1, 25] which study such CSPs in more general contexts.

[4]For a subset $S \subseteq V$, the hyperedge boundary $\partial^H_E(S)$ is the set of hyperedges crossing the cut (S, S^c) i.e., $\partial^E_H(S) = \{e \in E | e \cap S, e \cap S^c \neq \emptyset\}$.

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2 Overview

Our algorithmic result for SSVE is achieved via a known reduction to Hyper-SSE [30, 28], henceforth we shall focus our discussion on Hyper-SSE. Recall that the setting of Hyper-SSE is as follows: given a hypergraph $H = (V, E)$, the objective is to find a set $S \subseteq V$ of relative volume δ which minimizes the hyperedge expansion $\phi^{\mathsf{E}}_H(S)$ (as defined in (1.2)). As is usual, our approach to designing an algorithm for Hyper-SSE relies on the "SDP Relaxation + Gaussian Rounding" framework. However, towards designing the algorithm, we will encounter and address several fundamental issues which will guide our final approach. The rest of the section will consist of the following components.

- We first discuss the basic SDP relaxation and its integrality gap.
- We shall then sketch a relatively straightforward $\sqrt{\phi^{\mathsf{E}}_\delta \log(1/\delta)}$-algorithm for the "equals" version of small-set edge expansion (i.e., arity 2), and discuss the key bottlenecks towards extending this approach to the setting of hypergraphs of higher arities.
- Finally, we discuss our approach towards addressing these challenges and conclude by presenting a simplified version of our algorithm and its analysis.

2.1 The Basic SDP We begin by describing the basic SDP relaxation for the Hyper-SSE in SDP 1.

SDP 1.

$$\begin{array}{lcr} \textit{minimize} & \sum_{e\in E} \max_{i,j\in e} \|u_i - u_j\|^2 & \\ \textit{subject to} & \sum_{i\in V} \langle u_i, u_\emptyset\rangle = \delta n & \\ & \langle u_i, u_\emptyset\rangle = \|u_i\|^2 & \forall i \in V \\ & \|u_\emptyset\|^2 = 1. & \end{array}$$

In the above SDP, the objective is a convex vector relaxation of the hypergraph cut function. The variables i.e., the vectors $\{u_i\}_{i\in V}$ are intended to be $\{0,1\}$ indicators of a set in an integral solution, and $u_\emptyset$ is the purported "one" vector which aligns the SDP solution. The first constraint in the SDP relaxation is intended to control the cardinality of the set represented by the solution, and the second constraint is the vectorization of the Booleanity constraint $x^2 = x$ for $x \in \{0,1\}$. This relaxation and its instantiations to the setting of graphs (often strengthened with ℓ_2^2-triangle inequalities), have been used in most of the earlier work on small-set expansion problems [35, 9, 28]. While for constant arity (i.e. size of the largest hyperedge), the above SDP can yield optimal (up to constant factors and assuming SSEH) approximation results (e.g,. see [35] and [36]), in general the integrality gap is at least linear in the arity of the hypergraph, as stated in the following observation.

Observation 1. *[28] For hypergraphs with arity d, the integrality gap $\alpha_{\mathrm{Int}}(\delta, d, \epsilon)$ of the SDP 1 is at least $\Omega(\min\{1/\delta, d\})$.*

The integrality gap instance is simple to describe; it is a single hyperedge on d vertices. Here for $\delta = 1/d$ the optimal expansion is 1, since the hyperedge will always get cut. On other other hand, consider the following vector assignment. Let $u_\emptyset, \bar{z}_1, \ldots, \bar{z}_d$ be $(d+1)$-orthonormal vectors. For every $i \in [d]$, assign $u_i = \delta u_\emptyset + \sqrt{\delta - \delta^2}\bar{z}_i$. It is easy to verify that (a) this assignment is feasible and (b) this yields a value of $1/d$ for the objective. Observation 1 results in a situation where the integrality gap (denoted by α_{Int})[5] and SSEH based hardness factors (denoted by α_{SSE}) are *incomparable*. This leads us to ask the following which is another motivating question for this work: *Can we give an algorithm which achieves an approximation factor of $O(\max(\alpha_{\mathrm{SSE}}, \alpha_{\mathrm{Int}} \cdot \phi^{\mathsf{V}}_\delta))$?*

Remark 2. *We point out that the integrality curve $\alpha_{\mathrm{Int}}(\cdot)$ corresponds to that of the standard SDP relaxation described in SDP 1. In particular, $\alpha_{\mathrm{Int}}(\cdot)$ does not necessarily have to match the approximation curve $\alpha_{\mathrm{SSE}}(\cdot)$. On the other hand, it is clear that any rounding scheme for the above relaxation cannot have an approximation guarantee better than α_{Int}. Hence a restatement of the above question is that can we give achieve α_{SSE} approximation factor when $\alpha_{\mathrm{SSE}} < \alpha_{\mathrm{Int}} \cdot \phi^{\mathsf{V}}_\delta$?*

[5] Here $\alpha_{\mathrm{Int}} = \alpha_{\mathrm{Int}}(d, \delta, \epsilon)$ denotes the integrality gap of SDP 1 on δ-Hyper-SSE instances with maximum degree d and SDP value ϵ

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2.2 A $\sqrt{\phi^{\mathsf{E}}_\delta \log(1/\delta)}$-algorithm for SSE Here we sketch an algorithm for SSE which outputs $\delta|V|$-sized sets with edge expansion at most $\sqrt{\phi^{\mathsf{E}}_\delta \log(1/\delta)}$. Towards this, we begin by considering the instantiation of SDP 1 for the setting of graphs i.e,. $d = 2$:

SDP 2.

$$\begin{aligned}
\textit{minimize} \quad & \textstyle\sum_{(i,j)\in E} \|u_i - u_j\|^2 \\
\textit{subject to} \quad & \textstyle\sum_{i\in[n]} \langle u_i, u_\emptyset\rangle = \delta n \\
& \langle u_i, u_\emptyset\rangle = \|u_i\|^2 \qquad \forall i \in V \\
& \|u_\emptyset\|^2 = 1.
\end{aligned}$$

Idealized Analysis and a Gaussian Rounding Question. Given the SDP relaxation in SDP 2, the natural next step is to design and analyze a rounding algorithm which will use the SDP solution to round off a feasible integral solution – i.e., a δn-sized subset – with the objective that hyperedge boundary of the rounded solution is not too large in comparison to the SDP objective. This is the central step of the process and is key to determining the overall approximation guarantee of the algorithm. For the setting of graphs i.e, when the arity $d = 2$, designing optimal rounding schemes is a well understood process and is based on the following geometric principle: vertices whose corresponding vectors are far apart should be far more likely to be cut by the rounded off solution and vice versa. This is easily achieved by projecting the vectors along a randomly sampled Gaussian vector. Furthermore, in order to control the size of the set output by the algorithm, the partition of the vertices is guided by an appropriately chosen threshold. These with some additional technical considerations taken together yield the rounding scheme described in Figure 1.

Rounding for SSE

- For every $i \in V$, write $u_i = \mu_i u_\emptyset + z_i$, where $z_i \perp u_\emptyset$ is the component of u_i orthogonal to the one vector $u_\emptyset$.
- **Random Projection**. Sample $g \sim N(0,1)^k$ (where k is the dimension of the SDP solution) and for every $i \in V$, project $g_i := \left\langle g, \frac{z_i}{\|z_i\|}\right\rangle$.
- **Thresholding**. Construct $S \subseteq V$ by including every $i \in V$ for which if $g_i \le \Phi^{-1}(\delta)$.[a]
- Output set S.

[a] Here $\Phi : \mathbb{R} \to [0,1]$ is the Gaussian CDF function.

Figure 1: Basic Rounding Scheme

The correctness of the rounding scheme can be shown by establishing the following two steps: (i) the set returned is of size $\approx \delta|V|$ with high probability and (ii) the expected expansion of the rounded off solution is small. For establishing (i), one can easily see that the choice of the threshold ensures that for $i \in V$, marginally $g_i = \langle g, z_i/\|z_i\|\rangle$ is a standard Gaussian random variable, and hence

$$\Pr_{(x_i)_{i\in V}}\Big[x_i = 1\Big] = \Pr_{g\sim N(0,1)}\Big[g_i \le \Phi^{-1}(\delta)\Big] = \delta, \tag{2.3}$$

which implies that the expected size of the set rounded by the algorithm is $\delta|V|$. The more interesting component here is to analyze the expansion of the set S output by the algorithm (Figure 1). Equivalently, we want to bound the expected fraction of edges cut by S. It turns out that for a fixed edge $(i,j) \in E$, analyzing the probability of (i,j) getting cut reduces to the following Gaussian stability type question which asks "*what is the probability that two ρ-correlated Gaussians get separated by a halfspace of volume δ*"? This question and its variants have

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been studied in various contexts in the combinatorial optimization and graph partitioning problems in particular. Goemans and Williamson [17] first studied the above for the setting of $\delta = 0$; subsequent works such as [11] on UNIQUEGAMES and SSE study it for more general δ's. In particular, incorporating the bounds from [11] into the setting of our rounding scheme yields the following.

FACT 2.1. *Fix $i, j \in V$ and let $H_\delta = \{x \in \mathbb{R} | x \leq \Phi^{-1}(\delta)\}$ be the halfspace of Gaussian volume δ. Then,*

$$\Pr\left[\mathbb{1}_{H_\delta}(g_i) \neq \mathbb{1}_{H_\delta}(g_j)\right] \lesssim \|u_i - u_j\| \cdot \sqrt{\delta \log(1/\delta)}.$$

where $\lesssim$ hides constant multiplicative factors.

We point out that earlier works which employ Gaussian thresholding rounding study it for the setting where the random projection is done along the $\{u_i\}_{i\in V}$ vectors as opposed to the $\{z_i\}_{i\in V}$ vectors. Consequently, even for edges, one still needs derive the above bound specifically for our rounding scheme, although it follows in a straightforward manner using the techniques of [11]. Equipped with the above bound, (assuming G is ℓ-regular for simplicity) one can conclude the analysis of the approximation guarantee by bounding the expansion of the rounded set as:

$$\frac{\sum_{(i,j)} \|u_i - u_j\| \sqrt{\delta \log(1/\delta)}}{\delta \ell n} \leq \sqrt{\frac{|E|}{\ell n}} \cdot \sqrt{\frac{\sum_{(i,j)} \|u_i - u_j\|^2 \log(1/\delta)}{\delta \ell n}} \leq \sqrt{\phi^{\mathsf{E}}_\delta \log(1/\delta)}. \tag{2.4}$$

From the above discussion, it is easy to infer that a key bottleneck towards establishing analogous approximation guarantees for Hyper-SSE would be to extend Fact 2.1 to d-sets of correlated Gaussians (where $d > 2$). In fact, looking at the form of α_{SSE}, one might even be tempted to suggest that the following holds[6]:

QUESTION 1. *Given unit vectors $u_1, \ldots, u_d$, let $g_i := \left\langle g, \frac{z_i}{\|z_i\|}\right\rangle$ be the corresponding projected Gaussians (as in Figure 1). Furthermore, let $H_\delta \subset \mathbb{R}$ be the halfspace of Gaussian volume δ. Then*

$$\Pr_{(g_i)_{i\in[d]}}\left[\exists i, j \text{ s.t. } \mathbb{1}_{H_\delta}(g_i) \neq \mathbb{1}_{H_\delta}(g_j)\right] \overset{?}{\leq} \left\{\max_{i,j\in e} \|u_i - u_j\|\right\} \cdot \sqrt{\delta \log(1/\delta) \log(d)}, \tag{2.5}$$

If the above bound i.e, (2.5), held for arbitrary sets of vectors $u_1, \ldots, u_d$, then plugging in such a bound into the analysis described above would immediately yield an algorithm which outputs δ-weight sets with hyperedge expansion at most $\sqrt{\phi^{\mathsf{H}}_\delta \log(d) \log(1/\delta)}$ and we would be done. Unfortunately, it turns out that the above does not actually hold for arbitrary choices of $u_1, \ldots, u_d$, and the witness to that can again be constructed from the integrality gap example, as we discuss below.

Let $(u_i)_{i\in[d]}$ be the vector assignment for the integrality gap assignment i.e, $u_i = \delta u_\emptyset + \sqrt{\delta - \delta^2} \cdot \bar{z}_i$ for every $i \in V$ where recall that $\{\bar{z}_i\}$ are orthonormal vectors which are all orthogonal to $u_\emptyset$. Then observe that for $g_i = \langle g, \bar{z}_i\rangle$, the Gaussians $g_1, \ldots, g_d$ are *independent* standard Gaussian random variables and hence they separate the edge with probability at least $1 - (1-\delta)^d - \delta^d \geq \Omega(1)$ when $d = \Theta(1/\delta)$. On the other hand, the RHS of (2.5) is at most $\sqrt{\delta \cdot d^{-1} \cdot \log(1/\delta) \log(d)} \ll O(1)$ when δ is small, and hence the above assignment of vectors $\{v_i\}_{i\in[d]}$ violates (2.5). Therefore, understanding under what conditions one can establish (2.5) and how can one analyze hyperedges which for which the local distribution violates (2.5) is one of the key bottlenecks towards designing and analyzing an approximation algorithm for HYPER-SSE.

2.3 Our Approach: Nice edges and Gap edges. As described in the previous part, identifying robust characterizations of $(g_i)_{i\in[d]}$ from Question 1 under which one can expect to establish (2.5), and ways to handle edges for which the local distribution does not allow for (2.5), is one of the key challenges towards improving on existing bounds. Interestingly, it turns out that the fact that our example which violates (2.5) can be derived from the vector assignment of the integrality gap instance is not entirely coincidental. In fact, we can

[6]The intuition behind suggesting this bound is the following well-known Gaussian geometric fact (e.g., Proposition 8.6 [15]): for ϵ-small enough as a function of δ, d, if $g_1, \ldots, g_d$ were $(1-\epsilon)$-correlated copies of a standard Gaussian $g \sim N(0,1)$, then the RHS of (2.5) can be shown to be $\Theta(\sqrt{\epsilon \log d \log(1/\delta)})$.

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systematically show that collections of Gaussians (identified by the hyperedges in the hypergraph) that violate (2.5) have correlation structure which geometrically resemble the integrality gap vector assignment, in a somewhat approximate sense. This connection allows us to classify hyperedges as "nice" edges – for which (2.5) holds, and consequently we can argue Gaussian stability type approximation bounds, and "gap" edges – which resemble the integrality gap assignment, for which way pay a cost of $\tilde{O}(d \cdot \mathsf{Cost}(e))$ – where $\mathsf{Cost}(e)$ is the cost contributed by e to the objective.

Towards establishing the above, it is useful to adopt the probabilistic viewpoint of pseudo-distributions[7] corresponding to the vector solution. Given a hyperedge $e = \{1, \ldots, d\}$, let $(u_i)_{i\in[d]}$ be the corresponding set of vectors, for which we can write $u_i = \mu_i u_\emptyset + z_i$ for every $i \in [d]$, with z_i being the component of u_i orthogonal to the one-vector $u_\emptyset$. This is a useful decomposition since the various quantities derived from it can also be interpreted as probabilistic quantities related to the corresponding degree-2 pseudo-distribution. In particular, denoting the local variables associated with the pseudo-distribution as $X_1, \ldots, X_d$, we have that μ_i represents the (pseudo) probability of setting the variable X_i to 1, and $\langle z_i, z_j\rangle/\|z_i\|\|z_j\|$ represents the (pseudo) correlation between variables X_i and X_j. Consequently, note that the Gaussian ensemble $(g_i)_{i\in[d]}$ matches the correlation structure of the local variables $(X_i)_{i\in[d]}$ i.e, correlation between g_i and g_j is identical to that of X_i and X_j, for every $i, j \in [d]$. Furthermore, for any i, j, we have that $\|u_i - u_j\|^2$ represents (up to multiplicative factors) the probability of event $\{X_i \neq X_j\}$.

Now, from standard results in Gaussian noise stability[6] it is understood that one can establish (2.5) if the corresponding Gaussians are correlated enough. From a geometric perspective, this is expected to happen when the vectors v_i are close enough as a function of the biases μ_i. This intuition can be formalized using using the following probabilistic (and equivalently, geometric) inequality: for any pair of joinly distributed $\{0, 1\}$-random variables X_i, X_j, we have

$$\underbrace{\widetilde{\rho}(X_i, X_j) \geq 1 - \frac{\widetilde{\Pr}[X_i \neq X_j]}{\sigma(X_i)\sigma(X_j)}}_{\textit{Probabilistic}} \qquad\qquad \underbrace{\langle \bar{z}_i, \bar{z}_j\rangle \geq 1 - \frac{\|u_i - u_j\|^2}{\|z_i\|\|z_j\|}}_{\textit{Geometric}}$$

where $\widetilde{\rho}(\cdot, \cdot)$ denotes the correlation coefficient and $\sigma(\cdot)$ denotes the standard deviation. In particular, assuming $\mu_i = \delta \in (0, 1)$ for every $i \in [d]$ for simplicity, one can use the above inequality to derive a relationship between the minimum correlation between the Gaussian random variables and the contribution of the hyperedge to SDP objective.

$$\rho_e := \min_{i,j\in e} \rho(g_i, g_j) \geq 1 - \frac{\max_{i,j\in[d]} \|u_i - u_j\|^2}{\delta} \tag{2.6}$$

The above inequality immediately suggests that one can potentially analyze the performance of Gaussian rounding by considering the following extreme (but not exhaustive) cases. For brevity, denote $\nu_e := \max_{i,j\in[d]} \|u_i - u_j\|^2$. Then,

Case (i) "Nice^* Edges": Suppose $\nu_e \ll \delta$, then it follows that $\rho_e \sim 1$ i.e., we are in the almost correlated regime, where we can hope to show that Gaussian rounding yields a cut probability of $O(\sqrt{\nu_e} \cdot \sqrt{\delta \log(d) \log(1/\delta)})$ as suggested by (2.5).

Case (ii) "Gap^* Edges": Suppose $\nu_e \geq \delta$. Then, we can upper bound the probability of e getting cut by the probability of at least one of the variables being set to 1, which using (2.3) is at most $d \cdot \delta \leq d \cdot \nu_e$. Here, the contribution matches the multiplicative loss due to the integrality gap of the SDP.

The above classification into Nice^* and Gap^* edges via (2.6) forms the basis of our approach towards designing the algorithm for Theorem 1.1. However, note that the above only works in the "idealized setting" with the following unreasonable assumptions hold (i) all hyperedges can be classified into the idealized definitions of Nice^* and Gap^* edges, and (ii) locally the biases μ_i of hyperedges should all be identical to δ. Towards dealing with the above, we need to consider a modified rounding algorithm and additional pre-processing steps, and consider more relaxed notions of Nice and Gap edges which will allow us to smoothly interpolate between the two settings while covering all possible cases. Consequently, our modified rounding and relaxed notion of nice'ness imply that deriving (2.5) in the new setting becomes significantly more complicated.

[7]Informally, a degree-r pseudo-distribution is a collection of distributions on subsets of variables of sizes at most r that are locally consistent, see Section 3.2 of the full version for a more formal description.

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The Rounding Scheme, Gaussian Rounding Lemma, Additional Bottlenecks. A first step towards addressing issues (i) and (ii) outlined above is to use a (standard) alternative rounding scheme [12, 37] where instead of using a fixed bias for thresholding, we make the thresholds vertex sensitive. Formally, we replace the thresholding step in Figure 1 with the following.

▷ *For every $i \in V$, include $i \in S$ if $g_i \leq \Phi^{-1}(\mu_i)$,*

i.e., the threshold for vertex $i \in V$ is $\Phi^{-1}(\mu_i)$, where recall that $\Phi(\cdot)$ is the Gaussian CDF function. The varying thresholds allow us to define notions of Gap and Nice in the following more edge sensitive way. Fix a hyperedge $e = [d]$, and let $(v_i)_{i \in e}$ be the corresponding vectors from the vector solution of the SDP. As before, let us write $u_i = \mu_i u_\emptyset + z_i$ where $z_i \perp u_\emptyset$. Furthermore, for simplicity, we may assume $0 \leq \mu_d \leq \cdots \mu_1 \leq 1/2$ and as before, let ν_e denote the contribution of the hyperedge to the SDP objective. Then,

▷ *In the above setting, we say that the edge e is* Nice *if $\mu_1 \geq \nu_e \cdot \log(d) \log(1/\delta)^2$, and* Gap *otherwise.*

Given this setup, we can now derive the intended bounds for the probability of any hyperedge getting cut under the rounding scheme. As before, for Gap edges, we can bound the probability of the hyperedge getting cut by the probability of at least one of the variables being included in the set, which by a union bound and the definition of Gap edges is at most $d \cdot \mu_1 \leq \tilde{O}(d \cdot \nu_e)$. On the other hand, if the edge is Nice, then we can establish bounds on the cut probability using the following Gaussian rounding lemma which is the key technical contribution of this work.

Lemma 2.1. (Informal version of Lemma 5.1 in full version) *Suppose the hyperedge $e = [d]$ is* Nice. *Furthermore, suppose we have $\mu_i \geq \delta^{10}$ for every $i \in V$ and $|\mu_i - \mu_j| \leq \nu_e$ for every $i, j \in [d]$. Then, for $g \sim N(0,1)$, the ensemble of Gaussians $(g_i := \langle g, \bar{z}_i \rangle)_{i \in [d]}$ satisfy:*

$$\Pr_{g_1,\ldots,g_d}\left[\exists\ i,j \in [d],\ g_i \leq \Phi^{-1}(\mu_i) \ \ \&\ \ g_j > \Phi^{-1}(\mu_j)\right] \lesssim \sqrt{\mu_1 \nu_e \log(d) \log(1/\delta)}.$$

Establishing the above involves most of the technical work – this is mainly because, unlike the setting of Question 1, the above lemma applies in a much broader sense where the biases μ_i are not fixed to δ but can vary from vertex to vertex. The proof of the lemma involves a carefully combined application of several ingredients including two sided perturbation bounds of the Gaussian CDF function, large deviation inequalities for maximum of value of d-Gaussian random variables, and inequalities derived using the probabilistic (pseudo-distribution) interpretation of the vector solution. Additionally, we point out that the above lemma still doesn't apply unconditionally for Nice edges, and in particular, requires them to additionally satisfy the following:

- *δ-bounded away'ness*: We need the biases μ_i to be at least $\delta^{O(1)}$ – this is crucial in recovering the $\log(1/\delta)$-term in the approximation guarantee.
- *ℓ_1-bounded'ness.* We need the absolute differences of the biases μ_i, μ_j to be bounded in an ℓ_1-sense by the SDP cost ν_e.

The above conditions are again needed due to additional complications that arise due to the fact that the thresholds in our rounding scheme now are vertex sensitive. The former is addressed by a delicate pre-processing step which ensures that the modified vector solution satisfies the bounded'away ness condition without affecting the overall geometry of the vector solution. The ℓ_1-boundedness is ensured by introducing ℓ_1-constraints in the SDP relaxation, and then pre-deleting edges which violate the constraint. We elaborate on these in the next section.

2.4 The Final Algorithm and its Analysis

Our actual algorithm uses the following SoS lifting of the basic SDP described in Figure 1.

For ease of exposition, the objective and constraints in the above SDP are expressed in terms of constraints on the pseudo-distribution. Denoting $(X_i)_{i \in V}$ as the set of local variables, the objective is a sum of terms corresponding to hyperedges $e \in E$, where for every hyperedge $e \in E$, the corresponding term in the objective measures the maximum probability of separation $\Pr_{(X_i,X_j)\sim\mu_{ij}}[X_i \neq X_j]$ among all pairs $i, j \in e$ under the

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SDP Relaxation for Hyper-SSE

$$\min \quad \frac{1}{\delta n} \sum_{e\in E} w(e) \max_{i,j\in e} \widetilde{\Pr}_{(X_i,X_j)\sim\mu_{ij}} [X_i \neq X_j] \tag{2.7}$$

$$\mathsf{E}_{i\sim V_G} \widetilde{\Pr}_{X_i\sim\mu_i|X_S\leftarrow\alpha}\Big[X_i = 1\Big] = \delta \qquad \forall S : |S| \leq R-1, \alpha \in \{0,1\}^S \tag{2.8}$$

$$-\nu_e \leq \mu_i - \mu_j \leq \nu_e \qquad \forall i,j \in e, \forall e \in E \tag{2.9}$$

Figure 2: SoS SDP

pseudo-distribution. Note that this is equivalent to the corresponding vectorized term $\max_{i,j\in e} \|u_i - u_j\|^2$ (up to multiplicative constants), and hence the objective is identical to that of SDP 1 (up to scaling). In addition, as in SDP 1, we introduce constraints on the biases (2.8) to control the size of the set rounded off by the algorithm. Finally, we also include[8] ℓ_1-constraints for every hyperedge e which says that the biases μ_i, μ_j for vertices i, j inside the hyperedge e cannot deviate more than the maximum probability of an edge getting separated i.e, ν_e. Now we describe the rounding scheme in Figure 3 which happens in four steps.

Rounding for Hyper-SSE

Input. Let $\{\mu'_e\}_{e\in E}$ be the R-round optimal pseudo-distribution from solving the SDP relaxation from Figure 2.

I **Conditioning**. Choose a random subset S of size t and sample a labeling $X_S \leftarrow \alpha \sim \mu'_S$ from the local distribution on S. Let $\mu := \mu'|_{X_S\leftarrow\alpha}$ denote the resulting degree 2-SoS pseudo-distribution conditioned on $X_S \leftarrow \alpha$.

II **Edge Deletion**. Denoting $\nu_e = \max_{i,j\in e} \Pr_\mu [X_i \neq X_j]$, we delete all the hyperedges with $\nu_e \geq 1/10$.

III **Pre-processing**. Let $\mathbf{V} := \{v_i\}_{i\in V} \cup \{u_\emptyset\} \subset \mathbb{R}^\ell$ be the $\{\pm 1\}$-vector solution[a] corresponding to μ. Using $\mathbf{V}$, we construct a new vector solution $\{v'_i\}_{i\in V} \cup \{u_\emptyset\}$ as follows. Let $v_i = (1-\mu_i)u_\emptyset - 2z_i$ for every $i \in V$ as before, and let $\hat{z}$ be a unit vector orthogonal to $\mathbf{V}$. Then define new vectors $\{v'_i\}_{i\in V}$ as

$$v'_i = \frac{v_i - \theta \cdot \hat{z}}{\sqrt{1+\theta^2}},$$

where $\theta := \delta^{100}$.

IV **Gaussian Rounding**. Writing $v'_i = (1-2\mu'_i)u_\emptyset - 2z'_i$ for every $i \in V$, we round off an integral solution as follows: sample Gaussian vector $g \sim N(0,1)^\ell$ where ℓ is the ambient dimension of $\mathbf{V}'$ and construct $S \subset V$ as

$$S := \left\{ i \in V \middle| \left\langle g, \frac{z'_i}{\|z'_i\|} \right\rangle \leq \Phi^{-1}(\mu'_i) \right\}.$$

Output S.

[a]For technical reasons, it is easier to work with the $\{\pm 1\}$-version of the vector solution instead of $\{0,1\}$. However, the two are related by the identity $v_i : u_\emptyset - 2u_i$. Consequently, we have the equivalence $u_i = \mu_i u_\emptyset + z_i$ if and only if $v_i = (1-2\mu_i)u_\emptyset - 2z_i$.

Figure 3: HSSE Rounding

We give a brief sketch of the analysis for the above rounding algorithm.

[8]We point out that in the SoS lifting of the above SDP, (2.9) is enforced automatically, since it follows from the ℓ_2^2-triangle inequalities, which in turn is implied by the SoS constraints.

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Conditioning. The first step ensures that with high probability, the average correlation between the random variables is small. This ensures that the Gaussian rounding in step IV returns a set which is close to the intended size with large probability.

Edge Deletion. This step along with the ℓ_1-constraints ensures that for the surviving hyperedges, the "biases" (the bias of a vertex i is defined as μ_i) of vertices in a hyperedge are "close" to each other. A simple application of Markov's inequality shows the cost paid due to the deleted edges at most $10 \cdot \mathsf{Sdp\text{-}Cost}$.

Pre-processing. Here, given the degree-2 SoS solution $\{\mu_{S,\alpha}\}$, let $\{v_i\}_{i \in V} \cup \{v_\emptyset\}$ be the corresponding unique $\{\pm 1\}$-vector solution. As before, we express $v_i = (1 - \mu_i)u_\emptyset - 2z_i$ where $u_\emptyset$ is the one vector of the solution, and $\mu_i = \Pr_{X_i \sim \mu}[X_i = -1]$. From this, we construct a new vector solution $\{v_i'\}_{i \in V}$ given as

$$v_i' \stackrel{\text{def}}{=} \frac{v_i - \theta \hat{z}}{\sqrt{1 + \theta^2}} = \frac{(1 - 2\mu_i)u_\emptyset - 2z_i - \theta \hat{z}}{\sqrt{1 + \theta^2}}$$

where $\theta = \delta^{O(1)}$ is a constant depending only on δ and $\hat{z}$ is a unit vector that is orthogonal to $u_\emptyset, \{z_i\}_{i \in V}$. Although the resulting set of vectors need not constitute a degree-2 SoS solution, (for e.g., the Booleanity condition $x^2 = x$ need not be satisfied), we can still pretend that they come from a degree-2 pseudo-distribution and write them as $v_i' = (1 - 2\mu_i')u_\emptyset - 2z_i'$, where $u_\emptyset$ is the original one vector. A delicate analysis shows that they still *approximately* satisfy the properties of the original degree-2 solution, in addition to satisfying the δ-bounded away'ness condition needed to instantiate Lemma 2.1. We point the readers to Sections 4.2 and 4.3 of the full version for the full suite of properties we re-establish for the new vector solution.

Gaussian Rounding. Finally, we are in the setting where every surviving hyperedge $e \in E$ satisfies the δ-bounded away'ness condition, and the ℓ_1-bounded'ness condition required for instantiating Lemma 2.1. Now, as before, we shall now break the cut probability analysis into two cases. If e is Nice, then Lemma 2.1 applies and we can bound the probability of the hyperedge getting cut by $\sqrt{\nu_e \alpha_e \log d \log(1/\delta)}$ (where α_e is the bias farthest away from $\{0,1\}$). On the other hand, if the edge is a Gap edge, then we know $\mu_i \leq \tilde{O}(\nu_e)$ for every $i \in e$ and hence by a union bound, the probability of the edge getting cut is bounded by $\tilde{O}(d \cdot \nu_e)$. Combining these cut probability bounds with steps analogous to (2.4) yields the claimed approximation guarantee.

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